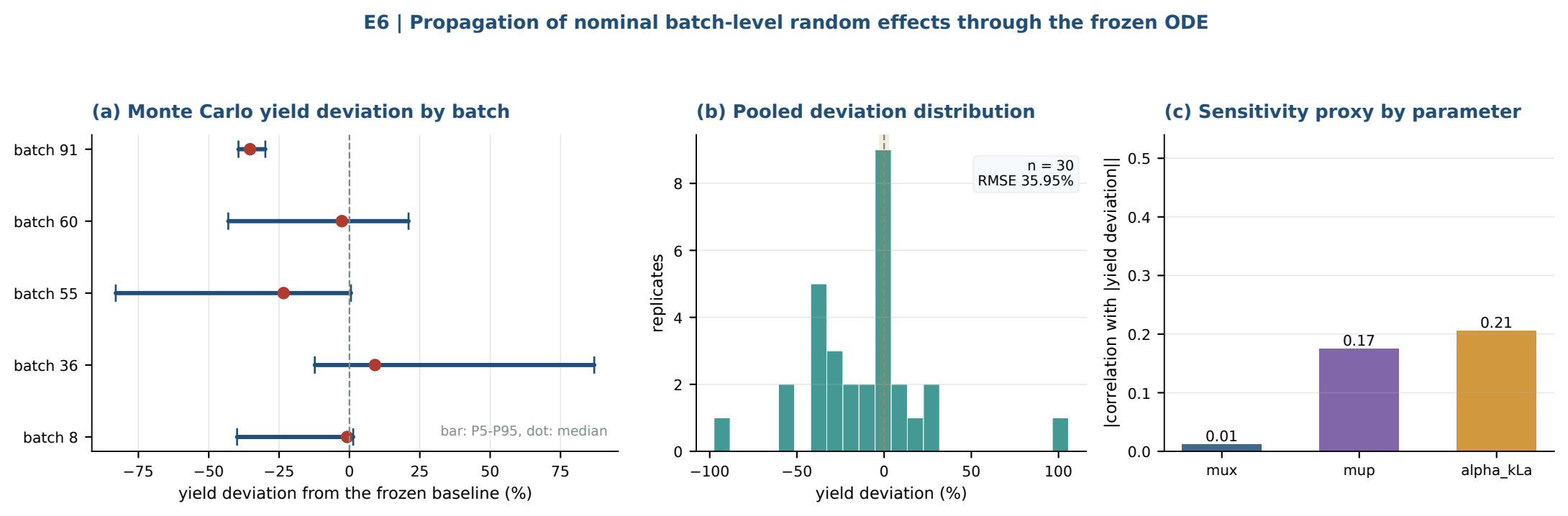
E6 | Propagation of nominal batch-level random effects through the frozen ODE
(a) Monte Carlo yield deviation by batch
batch 91
batch 60
batch 55
batch 36
batch 8
−75
−50
−25
0
25
50
75
yield deviation from the frozen baseline (%)
bar: P5-P95, dot: median
(b) Pooled deviation distribution
n = 30
RMSE 35.95%
replicates
0
2
4
6
8
−100
−50
0
50
100
yield deviation (%)
(c) Sensitivity proxy by parameter
|correlation with |yield deviation||
0.0
0.1
0.2
0.3
0.4
0.5
0.01
0.17
0.21
mux
mup
alpha_kLa